

Yash Rawal

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Professional Summary

Applied AI & Machine Learning Engineer serving a **1M+ user scale** by architecting enterprise agentic platforms, multi-agent systems, and Model Context Protocol (MCP) tool gateways across 120+ organizations. Specialized in LLM orchestration (LangGraph, ReAct loops, A2A protocol), SAP S/4HANA & SAP BTP AI integrations, enterprise RAG, and AgentOps evaluation control planes. Active contributor to Spring AI, bridging foundation models with Clean Core enterprise workflows.

Technical Skills

Languages & Backend: Python, Java, SQL, Spring Boot, FastAPI, PostgreSQL, SAP HANA, Data Engineering, REST APIs

Agentic Architectures: Agentic AI, Model Context Protocol (MCP), Agent-to-Agent (A2A), Multi-Agent Systems, Tool RAG, ReAct Loops, Clean Core

AI & ML Frameworks: LangChain, LangGraph, Spring AI, PyTorch, vLLM, LlamaIndex, GROBID, pdfplumber, Hugging Face

Platforms & Models: SAP AI Core, SAP Joule, SAP Build, Azure OpenAI, Amazon Bedrock, Google Vertex AI, Claude, GPT, Gemini, Meta Llama, Qwen

Infrastructure & Security: Docker, Kubernetes, Linux, CI/CD, LLM-as-Judge, Red Teaming, PII Sanitization (BIO NER), Prompt Injection Guardrails

Experience

Bigscal Technologies Pvt. Ltd.
AI/ML Engineer Oct 2024 - Present
Surat, India

- Deployed a production-scale agentic platform and multi-agent systems serving **1M+ active ERP users** across 120+ enterprise tenants on cloud infrastructure (Azure, AWS, GCP).
- Architected multi-agent workflows with LangChain and LangGraph, utilizing **MCP Server Tool Routing (Tool RAG)** and A2A communication with integrated guardrails.
- Designed a capability contract layer to execute nested profiles inside a single ReAct loop, reducing execution overhead by **3x** and token consumption by 70%.
- Implemented a hierarchical RAG pipeline indexing 10,000+ technical documents using GROBID, pdfplumber, and PostgreSQL/PGVector, achieving a **90%+ Recall@10** retrieval rate.
- Engineered an automated LLM evaluation and red-teaming control plane to log reasoning trajectories, scoring agent behaviors via **LLM-as-a-judge** and RLHF preference datasets.
- Built production guardrails against prompt injections and data leaks, integrating runtime tool validation and **automated PII redaction**.

Bigscal Technologies Pvt. Ltd.
AI/ML Engineer Intern Apr 2024 - Sep 2024
Surat, India

- Engineered clinical summarization pipelines for 116K+ medical records with 4-bit quantized Llama 3, and designed real-time WebSocket audio streaming for Indic ASR with **NeMo RNN-T and CTC decoding**.
- Researched audio digital signal processing (STFT, Mel-spectrograms, MFCC feature extraction via librosa/torchaudio) and CNN audio architectures (VGGish, ResNet), compiling custom Indic LJSpeech-format datasets.

Open Source & Projects

Modular Agent Harness & Memory System ("Brain") | Open Source
Principal Creator & Architect of an open-source plug-and-play agentic framework designed as modular building blocks for flexible enterprise workflows, implementing formal ReAct loops with step budgets, tactical pgvector tool routing, and dynamic MCP settings caching.
Java, Spring Boot, PostgreSQL, WebSockets

Spring AI (spring-projects/spring-ai) | Open Source
Merged PR #5711 to propagate reasoningContent from AWS Bedrock and OpenAI-compatible models, and resolved Converse API prompt-caching TTL latency consistency.
Java, Spring Boot, Python

Mini Language Model (~85M Parameter Architecture) | Personal Research
Pretrained an ~85M-parameter transformer to ~1.43 validation loss on TinyStories targeting **edge hardware deployment**; engineered FragmentStream chunked attention to eliminate $O(T^2)$ VRAM overhead and custom Triton kernels with KV caching for low-latency inference.
Python, PyTorch, CUDA, Triton, Tesla P100/T4

Education

Master of Computer Applications (Artificial Intelligence & Machine Learning) - Parul University 2022 - 2024

Certifications

Microsoft Certified: Azure AI Fundamentals (AI-900)	Microsoft
Microsoft Certified: Azure Fundamentals (AZ-900)	Microsoft